

SKILL WEEK-4

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Task 1: Process Synchronization and Child Process Monitoring Using waitpid()

```
saiviamsh@Varshini-PC:~/OSSP$ mkdir -p skill/week4/task1
saiviamsh@Varshini-PC:~/OSSP$ cd skill/week4/task1
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ nano task1.c
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ gcc task1.c -o T1
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ ./T1
Parent Process Started
Parent PID: 947

Parent is waiting for Child PID: 948

Child Process Started
Child PID: 948
Child is working...
Child Process Completed

Child terminated normally
Child Exit Status: 0
Parent Process Completed
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ |
```

Observation: The program successfully creates a child process using fork() and uses waitpid() to make the parent process wait for the child process to complete. The child process executes its task and terminates normally with exit status 0. After receiving the child's termination status, the parent process resumes and completes successfully. This demonstrates process synchronization and child process monitoring using waitpid().

Task 2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

```
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task1$ cd ~/OSSP
saivihamsh@Varshini-PC:~/OSSP$ mkdir -p skill/week4/task2
saivihamsh@Varshini-PC:~/OSSP$ cd skill/week4/task2
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ pwd
/home/madhulika/OSSP/skill/week4/task2
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ nano task2.c
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ./T2 Madhu
-bash: ./T2: No such file or directory
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ gcc task2.c -o T2
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ls
task2.c
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ./T2 Madhu
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/madhulika/AppData/Local/Programs/Python/Python313/Scripts:/mnt/c/Users/madhulika/AppData/Local/Programs/Python/Python313:/mnt/c/Users/madhulika/AppData/Local/Microsoft/Windows Apps:/mnt/c/Program Files/Java/jdk-21:/mnt/c/Users/madhulika/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/madhulika/AppData/Roaming/npm:/mnt/c/Users/madhulika/AppData/Local/Microsoft/WinGet/Packages/Gyan.FFMpeg_Microsoft.Winget.Source_8wekyb3d8bbwe/ffmpeg-8.1.2-full_build/bin:/snap/bin

Command 'Madhu' not found in PATH.
```

```
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ./T2 pwd
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/madhulika/AppData/Local/Programs/Python/Python313/Scripts:/mnt/c/Users/madhulika/AppData/Local/Programs/Python/Python313:/mnt/c/Users/madhulika/AppData/Local/Microsoft/Windows Apps:/mnt/c/Program Files/Java/jdk-21:/mnt/c/Users/madhulika/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/madhulika/AppData/Roaming/npm:/mnt/c/Users/madhulika/AppData/Local/Microsoft/WinGet/Packages/Gyan.FFMpeg_Microsoft.Winget.Source_8wekyb3d8bbwe/ffmpeg-8.1.2-full_build/bin:/snap/bin

File found: /usr/bin/pwd
Execute permission: YES
Command resolved successfully.
saivihamsh@Varshini-PC:~/OSSP/skill/week4/task2$
```

Observation: The program successfully retrieves the PATH environment variable and searches through its directories to locate the specified command. For the valid command pwd, it finds /usr/bin/pwd and verifies that execute permission is available. For the invalid command Madhu, it correctly reports that the command is not found in PATH. Thus, the program successfully demonstrates PATH parsing, executable searching, permission verification, and handling of missing commands.

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Task 1: To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

```
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ cd ~/OSSP
saiviamsh@Varshini-PC:~/OSSP$ mkdir -p skill/week4/task1
saiviamsh@Varshini-PC:~/OSSP$ cd skill/week4/task1
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ pwd
/home/madhulika/OSSP/skill/week4/task1
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ nano task1.c
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ ./T3
-bash: ./T3: No such file or directory
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ .T1
.T1: command not found
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ gcc task1.c -o T1
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ ./T1
Enter a string: ^C
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ pwd
/home/madhulika/OSSP/skill/week4/task1
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ ./T1
Enter a string: Madhu
Original : Madhu
Expanded : Madhu
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$ ./T1
Enter a string: hello $abc
Warning: Undefined variable: abc
Original : hello $abc
Expanded : hello
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task1$
```

Observation: The program successfully detects variable references in the input string and expands the variable values using environment variables. For a normal string such as Madhu, the input remains unchanged. When an undefined variable such as \$abc is entered, the program displays a warning and removes the undefined variable from the expanded output. Thus, the program successfully demonstrates variable detection, value expansion, and handling of undefined variables.

Task 2: To Identify Built-ins, Create Dispatch Table, Execute In-Process Built-ins, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ gcc task2.c -o T2
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ./T2
myshell> exit
Exiting shell...
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ cd ~/OSSP
saiviamsh@Varshini-PC:~/OSSP$ cd ~/OSSP/skill/week4/task2
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ pwd
/home/madhulika/OSSP/skill/week4/task2
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ nano task2.c
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ gcc task2.c -o T2
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ls
saiviamsh@Varshini-PC task2.c
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$ ./T2
myshell> echo Hello World
Hello World
myshell> cd/tmp
Command not found: cd/tmp
myshell> abc
Command not found: abc
myshell> cd /tmp
myshell> pwd
/tmp
myshell> exit
Exiting shell...
saiviamsh@Varshini-PC:~/OSSP/skill/week4/task2$
```

Observation: The program successfully identifies and executes built-in commands such as pwd, echo, cd, and exit using command dispatch logic. The cd command changes the current working directory and the change is maintained when pwd is executed again. Invalid commands are detected and reported as “Command not found.” Thus, the program successfully demonstrates built-in command identification, in-process execution, state maintenance, invalid-command handling, and dispatch logic testing.